

Enterprise SOC Lab

Active Directory Domain Services - Master Implementation Record

This is the consolidated technical record of the AD-DS implementation for the enterprise SOC lab. It combines two bodies of work: the original deployment carried out and documented prior to handover (Domain Controller build, AD-integrated DNS, and initial Linux domain-join procedure), and the extended work completed afterward (organizational structure, access-delegation groups, sudo authorization policy, and integration of Active Directory event data into the SOC's ELK-based SIEM with working Kibana detection rules). Every fact, command, and issue below reflects what was actually configured and verified, not a generic AD-DS how-to.

Domain	Detail
AD Domain / Kerberos Realm	kpk.local / KPK.LOCAL
Domain Controller	KPK-DC01 - Windows Server 2022 (AD DS, DNS, Global Catalog)
Linux hosts joined	SOC-Core, VM-MISP-01, VM-IDS-01 (Ubuntu 22.04.5 LTS)
Domain SID prefix	S-1-5-21-379509529-562884115-830820509
SIEM	Elasticsearch + Logstash + Kibana 8.11.4, Kafka-buffered pipeline, on SOC-Core

Part A - Pre-Handover Deployment (as documented and submitted)

The following summarizes the deployment carried out and formally documented before this engagement began, per the submitted "Active Directory Domain Services (AD DS) Implementation Report" and its three-phase deployment log. It is reproduced here in condensed form for a single point of reference; Part B covers everything built and verified afterward.

A.1 Objective and Scope

Establish a centralized identity and authentication service for the SOC lab via a Windows Server 2022 Domain Controller, providing centralized user/computer management, authentication, DNS, and a foundation for later Linux security-infrastructure integration.

A.2 Network Architecture (as designed)

Segment	CIDR	Hosts
DMZ	192.168.100.0/24	Web Server (.10), Honeypot (.100), future AWS web server
Internal Security Network	192.168.150.0/24	MISP (.10), SOC-Core (.20), IDS (.30), Domain Controller (.40)
Edge Gateway (AWS)	n/a	Ubuntu-based WireGuard VPN hub, routing, NAT, planned nftables

A.3 Domain Controller Build (Phase 1)

- OS: Windows Server 2022. Hostname: KPK-DC01. Domain: kpk.local, deployed as a new forest/root domain.
- Static IP assigned prior to promotion: 192.168.150.40 /24, DNS pointed at 127.0.0.1 (self).
- Roles installed: Active Directory Domain Services, DNS Server; server then promoted to Domain Controller.
- Verified post-promotion: Domain Controller active, Global Catalog enabled, all FSMO roles assigned, via Get-ADDomainController / Get-ADDomain / Get-WindowsFeature.

A.4 DNS Configuration (Phase 1 and 2)

AD-integrated DNS with forward lookup zones kpk.local and _msdcs.kpk.local. Static A records created for all enterprise hosts via Server Manager -> DNS -> New Host:

Host	FQDN	IP
KPK-DC01	kpk-dc01.kpk.local	192.168.150.40
misp	misp.kpk.local	192.168.150.10
soc-core	soc-core.kpk.local	192.168.150.20
ids	ids.kpk.local	192.168.150.30
web	web.kpk.local	192.168.100.10
honeypot	honeypot.kpk.local	192.168.100.100

SRV service-discovery records validated at both _ldap._tcp.dc._msdcs.kpk.local and _kerberos._tcp.dc._msdcs.kpk.local, both correctly advertising KPK-DC01 -> 192.168.150.40, using Resolve-DnsName -Type SRV.

A.5 Major Issue Encountered Pre-Handover: Multi-Homed DNS Advertisement

LDAP and Kerberos initially advertised incorrect IPs - first 192.168.74.176, later also 10.35.119.1 - instead of 192.168.150.40.

Root cause: the Domain Controller had multiple active network interfaces (WireGuard, a VMware virtual adapter, and a USB tethering interface). Netlogon automatically registered every eligible interface in AD DNS, so service records advertised inconsistent, sometimes non-routable IPs.

Resolution: updated the DC's A record, disabled DNS registration on the non-enterprise interfaces, flushed and re-registered DNS, and restarted Netlogon:

```
ipconfig /flushdns
ipconfig /registerdns
Restart-Service Netlogon
nltest /dsregdns
```

Final result: LDAP and Kerberos advertised only 192.168.150.40. This is flagged in the pre-handover report itself as a key technical finding - multi-homed Domain Controllers will register every eligible interface unless explicitly restricted, and only the enterprise-facing interface should be permitted to register in AD DNS. (Note: a related, distinct recurrence of this same class of issue was found and fixed again during the post-handover work - see Part B, section B.5, issue B.5.4.)

A.6 Linux Integration Procedure (Phase 3, as planned)

The pre-handover documentation laid out the intended Linux domain-join procedure for SOC-Core, MISP, and IDS (HoneyPot intentionally excluded to keep it isolated from enterprise identity; Web Server optional): configure DNS to 192.168.150.40, install realmd/sss/adcli/krb5-user, discover and join the domain, enable pam-auth-update mkhomedir, and validate via kinit/klist/id/getent. This plan was carried out in practice during the post-handover engagement - see Part B, section B.3 for the actual commands run, the real output observed, and every deviation from this plan that came up in execution.

Part B - Post-Handover Implementation (this engagement)

Everything below was configured and verified directly during this engagement, building on the Domain Controller and DNS foundation described in Part A.

B.1 Active Directory Structure Built

Organizational Units - created under DC=kpk,DC=local:

OU	Purpose
Domain Controllers	Built-in; holds KPK-DC01
Servers	Parent OU for server computer objects
Servers -> Windows Servers	Reserved for future Windows member servers
Servers -> Linux Servers	Holds SOC-CORE, VM-MISP-01, VM-IDS-01 computer objects
Groups	Holds all nine security groups
Service Accounts	Holds svc-misp, svc-logstash, svc-kibana, svc-suricata
SOC Users	Holds human accounts - named to avoid colliding with the built-in Users container (see issue B.5.1)

Security groups - nine total, in two categories:

Group	Category	Membership
SOC-Admins	Role	soc.admin1
SOC-Analysts	Role	soc.analyst1
ThreatIntel	Role	Empty - reserved
IR-Team	Role	Empty - reserved
ELK-Admins	Role	svc-logstash, svc-kibana
MISP-Users	Role	svc-misp
Linux-Sudo-Admins	Access-delegation	soc.admin1
RDP-Access	Access-delegation	Empty - reserved
Read-Only-Analysts	Access-delegation	Empty - reserved

Role groups answer "what team is this identity on"; access-delegation groups answer "what specific privilege does this grant" - kept as a separate axis deliberately, so privilege can be changed without touching team membership and vice versa.

Service accounts - PasswordNeverExpires=True, CannotChangePassword=True:

Account	Consumer	SID suffix
svc-misp	MISP application (planned LDAP bind account)	...-2112
svc-logstash	Logstash service identity	...-2113
svc-kibana	Kibana service identity	...-2114
svc-suricata	Suricata service identity	...-2115

Human test accounts - both in SOC Users:

Account	Purpose	Groups	Linux identity
soc.analyst1	Zero-privilege authentication baseline	SOC-Analysts	uid=592002117 gid=592000513(domain users)
soc.admin1	Privileged test account for sudo delegation	SOC-Admins, Linux-Sudo-Admins	Created in SOC Users; sudo verified working on SOC-Core

B.2 DNS Completion

Part A covered forward DNS. The following was added and verified during this engagement:

Item	Detail
Reverse zones	100.168.192.in-addr.arpa (DMZ) and 150.168.192.in-addr.arpa (Internal) - found already present; PTR records added for all hosts
PTR records	web(.10), honeypot(.100) in the 100.x zone; misp(.10), soc-core(.20), ids(.30), kpk-dc01(.40) in the 150.x zone
SRV validation (plain-domain form)	_ldap._tcp.kpk.local and _kerberos._tcp.kpk.local both confirmed via dig, resolving to kpk-dc01.kpk.local / 192.168.150.40

B.3 Linux Domain-Join - Actual Execution

Package set installed on all three hosts: realmd, sssd, sssd-tools, libnss-sss, libpam-sss, adcli, samba-common-bin, oddjob, oddjob-mkhomedir, packagekit, krb5-user.

```
sudo realm discover kpk.local
sudo realm join --user=Administrator \
  --computer-ou="OU=Linux Servers,OU=Servers,DC=kpk,DC=local" kpk.local
```

Final working sssd.conf domain section (all three hosts, after fixes described in B.5):

```
[domain/kpk.local]
default_shell = /bin/bash
krb5_store_password_if_offline = True
cache_credentials = True
krb5_realm = KPK.LOCAL
realmd_tags = manages-system joined-with-adcli
id_provider = ad
fallback_homedir = /home/%u@%d
ad_domain = kpk.local
use_fully_qualified_names = False
ldap_id_mapping = True
access_provider = simple
simple_allow_users = soc.analyst1
dyndns_update = False
```

Host	Computer object DN	Result
SOC-Core	CN=SOC-CORE,OU=Linux Servers,OU=Servers,DC=kpk,DC=local	Fully verified - Kerberos (kinit/klist), login, home dir, sudo all confirmed
VM-MISP-01	CN=VM-MISP-01,OU=Linux Servers,OU=Servers,DC=kpk,DC=local	Verified - login, home dir, dyndns errors resolved; sudo login for soc.admin1 failed once on a stale password, not yet re-confirmed
VM-IDS-01	Joined via consolidated script (not independently queried via Get-ADComputer)	Fully verified - clean join, login, automatic Kerberos ticket acquisition confirmed for soc.analyst1

B.4 Linux Sudo / Authorization Policy

Domain join and SSSD establish authentication only. Linux authorization (sudo) is separate and local, and was configured explicitly:

```
echo "%linux-sudo-admins ALL=(ALL) ALL" | sudo tee /etc/sudoers.d/ad-admins
sudo chmod 0440 /etc/sudoers.d/ad-admins # required - sudoers.d files rejected otherwise
sudo visudo -c                          # syntax + permission validation
sudo realm permit soc.admin1@kpk.local   # per-host login allowlist, separate from AD
group membership
```

Applied and validated (visudo -c: parsed OK) on all three hosts. End-to-end proof on SOC-Core: soc.admin1 logged in, sudo pwd succeeded (root access confirmed); soc.analyst1 remained correctly denied sudo ("is not in the sudoers file"), confirming the rule is properly scoped to the Linux-Sudo-Admins group and not accidentally open to all users.

B.5 Troubleshooting Log - Identity, DNS, and Access Issues

In the order encountered.

B.5.1 OU creation collision on "Users"

Symptom	New-ADOrganizationalUnit failed for an OU named "Users" at domain root.
Root cause	AD auto-creates default containers (Users, Computers, Builtin) at domain root; these occupy the namespace but cannot host Group Policy.
Resolution	Created the human-user OU as "SOC Users" instead.

B.5.2 Reverse zone creation reported "already exists"

Symptom	Add-DnsServerPrimaryZone failed for both reverse zones.
Root cause	Both zones had already been created (AD-integrated), likely alongside earlier forward-zone work.
Resolution	Verified existing zones via Get-DnsServerZone; proceeded directly to PTR record creation.

B.5.3 Duplicate PTR record for the Domain Controller

Symptom	Two PTR records existed for .40 - auto-registered plus a manually added duplicate.
Root cause	The DC self-registers its own PTR record on a timer; a second manual record was added on top.
Resolution	Removed the manual duplicate, kept the DC's self-managed record.

B.5.4 Duplicate/stray A record for the DC's own hostname (recurrence of the multi-homed pattern from Part A.5)

Symptom	kpk-dc01.kpk.local resolved to two A records: the correct 192.168.150.40 and an unrelated 192.168.74.152.
Root cause	The DC has a second, DHCP-leased NIC (Ethernet1) on the dynamic transport network; Windows auto-registers DNS for every interface by default - the same underlying behavior documented in Part A.5, recurring on a different interface after the earlier fix.
Resolution	Disabled DNS self-registration on Ethernet1 and removed the stray record.

```
Set-DnsClient -InterfaceAlias "Ethernet1" -RegisterThisConnectionsAddress $false
Remove-DnsServerResourceRecord -ZoneName "kpk.local" -Name "kpk-dc01" \
  -RRType A -RecordData "192.168.74.152" -Force
```

B.5.5 realm permit rejected the target username

Symptom	realm permit soc.analyst1 failed: login format mismatch.
Root cause	At that point realmd's login format was still the fully-qualified default (%U@kpk.local).
Resolution	Re-issued using the fully-qualified form to match the format active at that moment.

B.5.6 "No such user" after a successful domain join

Symptom	getent passwd / id returned nothing for soc.analyst1 despite successful join and permit.
Root cause	use_fully_qualified_names = True in [domain/kpk.local] overrides realmd's login-format setting; an earlier attempted fix had also been placed in the wrong config section ([sssd] instead of [domain/kpk.local]), where SSSD silently ignores it.
Resolution	Removed the misplaced setting; set use_fully_qualified_names = False in the correct section; cleared the SSSD cache and restarted the service.

B.5.7 Terminal output corruption masked two failed commands

Symptom	A forgotten background tail -f job interleaved with subsequent command output, making a config fix and service restart look successful when neither had actually run.
Root cause	Overlapping stdout from a stray background job obscured real command results.
Resolution	Killed the background job; re-ran each command individually with an explicit verification step after each one.

B.5.8 Password change during su failed ("Old password not accepted")

Symptom	su detected an expired password, prompted for change, then failed with an authentication token error.
Root cause	Prompt-sequence entry error (re-typing the current password incorrectly). Ruled out as an infrastructure fault: TCP 464 (kpasswd, the distinct AD password-change protocol/port, separate from Kerberos ticket-granting on port 88) was confirmed reachable, and a direct kpasswd change succeeded immediately.
Resolution	Changed the password via kpasswd directly; su then authenticated normally.

B.5.9 Home directory not created on first login

Symptom	su succeeded but warned the home directory didn't exist and landed the session in the wrong directory.
Root cause	oddjob-mkhomedir was installed and oddjobd was running, but the PAM hook (pam_mkhomedir.so) was never registered in /etc/pam.d/common-session.
Resolution	Enabled the hook via pam-auth-update --enable mkhomedir.

B.5.10 Recurring SSSD GSSAPI/TSIG errors

Symptom	SSSD logs repeatedly showed "tkey query failed", "TSIG error", "update failed: REFUSED" on SOC-Core and MISP.
Root cause	SSSD's dynamic DNS update feature (dyndns_update, default on) tried to refresh its own A record via the joined machine account's Kerberos identity, but those records were created manually under Administrator, so the machine account lacked write permission - a correct security refusal, not a bug.
Resolution	Disabled the feature (dyndns_update = False) on SOC-Core and MISP, and baked it into the join script for IDS from the start - IDS joined with zero such errors.

B.5.11 sudo denied for the AD test user (expected)

Symptom	soc.analyst1 correctly denied sudo on IDS with "is not in the sudoers file".
Root cause	Not a fault - authentication and authorization are separate; AD group membership grants no Linux privilege automatically.
Resolution	No fix needed; confirms correct, intentional behavior.

B.5.12 sudoers.d file rejected: bad permissions

Symptom	visudo -c reported "/etc/sudoers.d/ad-admins: bad permissions, should be mode 0440" after creating it with sudo tee.
Root cause	sudo tee creates files with default (looser) permissions; sudoers.d strictly requires 0440 as a deliberate anti-privilege-escalation guard.
Resolution	Ran chmod 0440 on the file, then re-validated with visudo -c on all three hosts.

B.5.13 soc.admin1 authentication failure on MISP

Symptom	su - soc.admin1 failed with "Authentication failure" on MISP, while the same account and sudoers rule worked cleanly on SOC-Core.
Root cause	The account was created with ChangePasswordAtLogon and had not yet had its password set via a successful change flow at the time of the MISP attempt (same underlying pattern as B.5.8).
Resolution	Not yet independently re-confirmed as resolved on MISP - remains an open item (see Part C).

B.6 SIEM Integration - Shipping AD-DS Events to ELK

B.6.1 Existing pipeline architecture (discovered, not built from scratch)

SOC-Core already ran a mature, multi-stage Logstash pipeline, defined in `~/project/logstash/pipeline/` and orchestrated via `pipelines.yml`:

Stage	File	Role
01 - Ingest	01-ingest.conf	Beats listener on port 5044; classifies incoming events by source (CloudTrail, CI/CD, Auditbeat, Suricata, nginx, auth.log, DNS query logs) and normalizes to ECS fields; publishes to Kafka topic <code>soc-events</code>
02 - Indexer	02-indexer.conf	Consumes from Kafka; enriches Suricata DNS events against MISP IOC data; routes parse failures to <code>soc-failed-*</code> , everything else to <code>soc-<dataset>-YYYY.MM.dd</code> in Elasticsearch
03/05 - MISP	03-misp.conf, 05-misp-ioc.conf	Threat-intel enrichment pipelines

The pipeline directory is mounted read-only inside the Logstash container; edits were made on the SOC-Core host filesystem and picked up via `docker restart logstash`. Elasticsearch runs with `xpack.security.enabled=true`, requiring the elastic user for all API calls.

B.6.2 Winlogbeat on KPK-DC01

Found to be already installed and shipping to SOC-Core:5044 prior to this engagement's SIEM work beginning - confirmed via live Logstash logs showing `agent.type: winlogbeat` traffic from KPK-DC01 before any new configuration was added. Collecting the Security, System, and Application channels.

B.6.3 New Logstash filter branch added for Winlogbeat

A new `else-if` branch was added to `01-ingest.conf` (inserted via a purpose-written Python script with automatic timestamped backup and brace-balance validation, matching the file's existing backup convention), detecting `agent.type == "winlogbeat"` and:

- Routes by `winlog.channel` into distinct datasets: `windows.security`, `windows.system`, `windows.application`, `windows.other` - landing in separate Elasticsearch indices.
- Maps the following Windows Security event IDs to ECS `event.category` / `event.type` / `event.action` / `event.outcome`: 4624/4625 (logon success/failure), 4634 (logoff), 4720/4725/4726 (user account created/disabled/deleted), 4728/4732/4756 (group member added), 4729/4733/4757 (group member removed), 4738 (user account changed), 4740/4767 (account locked/unlocked), 4771 (Kerberos pre-auth failed), 4776 (credential validation), 5136 (directory object modified).
- Copies `TargetUserName`, `SubjectUserName`, and `IpAddress` from `winlog.event_data` into the standard ECS `user.target.name`, `user.name`, and `source.ip` fields.

B.6.4 Two real bugs found and fixed during verification

Bug 1 Event-ID comparisons silently never matched

Symptom	event.category/action/outcome were never populated for Winlogbeat events, even though dataset routing (windows.security, etc.) worked correctly.
Root cause	winlog.event_id arrives as a JSON string ("4634"), but the filter compared it to a bare integer (4634). Logstash's == does not coerce types, so every event-ID branch evaluated false while the channel-based string comparisons worked fine.
Resolution	Quoted every event_id comparison as a string via a scoped sed replacement; verified against a live document afterward showing event.category: "authentication" correctly populated.

```
sed -i -E 's/[winlog\]\[event_id\] == ([0-9]+)/[winlog][event_id] == "\1"/g' \
logstash/pipeline/01-ingest.conf
```

Bug 2 event.action returned as a two-element array

Symptom	A retrieved document showed event.action: ["Logoff", "logoff"] instead of a single string.
Root cause	Winlogbeat has built-in ECS awareness for some well-known Security events and had already set event.action = "Logoff" before the event reached the filter. The custom mutate used add_field, which appends to an existing field rather than overwriting it, producing an array.
Resolution	Changed every add_field to replace within the Winlogbeat block (scoped Python edit, preserving other pipeline sections untouched) - replace deterministically overwrites regardless of any prior value. Verified: event.action returned as a single clean string afterward.

B.6.5 End-to-end verification performed

```
GET soc-windows.security-*/_search { match: { winlog.event_id: "4634" } }
-> event.dataset: windows.security
-> event.category: authentication    event.type: end
-> event.action: logoff              event.outcome: success
```

Confirmed correct, single-valued, ECS-consistent fields flowing all the way from a real Windows Security event through Winlogbeat, Logstash (both stages), Kafka, and into Elasticsearch.

B.7 Kibana Detection Rules

Rule type used: Elasticsearch query ("Custom Query Rule"), KQL/Lucene mode, against the "SOC Logs" Data View. Note: on inspection, the environment's 21 pre-existing rules (e.g. "AWS CloudTrail Logging Disabled", "AWS Security Group Modified") were all found to have 0 conditions configured - functioning shells that run and report "Succeeded" without matching or alerting on anything. The two rules below are believed to be the first genuinely working custom rules in this environment.

B.7.1 Rule: AD Brute Force - Repeated Logon Kerberos Failures

Setting	Value
Query	winlog.event_id: "4625" or winlog.event_id: "4771"
Data View	SOC Logs
Group by	Top 5, source.ip.keyword
Threshold	IS ABOVE 5 (temporarily lowered to 3 during live-fire testing)
Time window	5 minutes
Check every	1 minute
Status at end of session	Saved; live-fire test in progress - deliberate failed logons generated on KPK-DC01 and confirmed present in Elasticsearch (5 matching documents), but the rule's Alerts tab had not yet been confirmed to show a fired alert before this record was produced

B.7.2 Rule: AD Privileged Group Membership Changed

Setting	Value
Query	winlog.event_id: "4728" or winlog.event_id: "4732" or winlog.event_id: "4756"
Data View	SOC Logs
Group by	None - every occurrence should alert individually
Threshold	IS ABOVE OR EQUALS 1
Time window	1 minute
Check every	1 minute
Status at end of session	Planned/specified; not yet built in the Kibana UI at the time of this record

Rationale: rule 1 catches distributed/repeated authentication failure patterns per source (a brute-force signature); rule 2 is deliberately zero-tolerance, since privileged group membership changes should be rare, deliberate events worth surfacing immediately rather than only after a pattern repeats.

Part C - Consolidated Status and Open Items

Item	Status
Domain Controller build (Windows Server 2022, kpk.local)	Complete (pre-handover)
AD-integrated forward DNS + SRV validation	Complete (pre-handover)
Multi-homed DNS advertisement issue (original, 3 interfaces)	Resolved (pre-handover)
Reverse DNS zones + PTR records	Complete (this engagement)
Recurrence of multi-homed DNS issue (DC transport NIC)	Resolved (this engagement)
AD organizational structure - 7 OUs, 9 groups, 4 service accounts, 2 test users	Complete
SOC-Core, MISP, IDS domain join	Complete, all three verified
Linux sudo authorization via Linux-Sudo-Admins group	Complete and proven on SOC-Core
soc.admin1 password / sudo verification on MISP and IDS specifically	Open - MISP hit an authentication failure not yet re-confirmed fixed; IDS/MISP not individually re-tested after SOC-Core succeeded
Explicit kinit/klist for soc.analyst1 on SOC-Core and MISP (vs. Administrator)	Open - only confirmed explicitly on IDS
Winlogbeat -> Logstash -> Kafka -> Elasticsearch pipeline for AD/Kerberos events	Complete and verified with real data
Kibana rule: AD Brute Force	Saved; live-fire confirmation of an actual fired alert still pending
Kibana rule: AD Privileged Group Change	Specified, not yet built in the UI
MISP application's own LDAP integration (Docker container, distinct from host login)	Not started
Windows Event Forwarding to SOC-Core (native, separate from Winlogbeat)	Not started - Winlogbeat serves the equivalent purpose currently
nftables firewall on the VPN gateway	Not started
Group Policy Objects (password/audit policy)	Not started - discussed as concepts only

Recommended immediate next steps

- Re-verify soc.admin1's password/login state on MISP and IDS specifically (kpasswd, then su + sudo test).
- Confirm the Brute Force rule actually produces a fired alert in its Alerts tab, not just a positive Test Query preview.
- Build the Privileged Group Change rule in the Kibana UI using the specification in B.7.2.
- Close the Kerberos verification gap for soc.analyst1 on SOC-Core and MISP.
- Proceed to MISP LDAP integration and/or nftables once the above are closed out.